



SWEET Call X-2020: SWEET EDGE

Deliverable report

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Deliverable name	Model and plan for regional integration
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Summary

The interplay of new renewable installations (wind and solar PV) with existing hydropower needs to be investigated not only at a national level but down to smaller landscape units. Based on an existing optimization model (OREES), the demand – production balancing can be assessed at high spatial resolution. The methodology has been used for a first assessment of electricity transfer between landscape units under a scenario of new renewable production of 25 TWh, which can come from wind and solar PV. The scenarios have been calculated for economic optimization, which leads to a predominance of solar PV and for import minimization, which leads to a predominance of wind installations. In both scenarios, net transfer from the Alps to the cities is an important result, especially in the winter and amounts to at least twice the total quantity that needs to be imported. This first demonstration case is made with simplified restrictions for PV and wind placement, which can be refined in future.



1 Introduction

The leading idea of the EDGE project is to promote de-centralized installations of renewable energy sources such that specific landscape characteristics (Alps, Midlands, Cities) are optimally respected and such that local to regional demand is best served. Generally, in mountain regions, more energy is produced than consumed. However, the production does not (always) qualify as de-centralized production, at least not for big hydropower installations. Thus, new potential installations of solar PV and wind need to be assessed with respect to de-centralized demand. This deliverable provides a methodology, which allows to do this for larger and smaller landscape units. As a use case and first example, the half-yearly (summer and winter) net electricity transfer between nine landscape units is presented.

2 Deliverable content

Based on the OREES model (Dujardin et al., 2021), we have developed a methodology to assess demand – production balancing in the nine detailed EDGE landscape units. This can also be aggregated to the original landscape units “Alps”, “Midlands” and “Cities”. We used the new methodology to explore the electricity balancing between the 9 EDGE landscape units. Based on the model intercomparison study (Heinisch et al., 2023) the following scenarios were simulated:

- Scenario A (Figure 1): MIP HighRES35 scenario. Optimal investment in PV and wind power to reach 25 TWh/year, based on estimated demand profile and investment cost for 2035 (0.87 EUR/kWp for PV and 1.25 EUR/kWp for wind power). The optimization suggests an optimal portfolio composed of 13.6 GWp of PV panels and 1.14 GWp of wind turbines.
- Scenario B (Figure 2): Optimal portfolio of PV and wind power to decrease the amount of necessary import, under the same demand as scenario A, but without considering investment costs. Here, in order to compare scenario A and B, 14.74 GWp of PV panels and wind turbines are installed. The optimization suggests an optimal distribution composed of 1.98 GWp of PV panels and 12.76 GWp of wind turbines, with a total generation of 27.3 TWh/year.

OREES provides the optimal installation maps on a 1.6 x 2.3 km² grid for PV and 1.1 x 1.1 km² for wind power. Moreover, the model provides spatially distributed time series of generation from hydropower (run-of-river, storage hydropower and pumped-hydropower), as well as the necessary imports and exports of electricity with the neighbouring countries. Hydropower has a lot of spatial details not only for the large plants such as to make sure that regional integration of new renewables and its interaction with hydropower is well represented. As an example, Figure 1 and Figure 2 show a summary of the electricity balancing between the 9 EDGE landscape units and the necessary imports and exports for the two scenarios and for the winter and summer seasons.

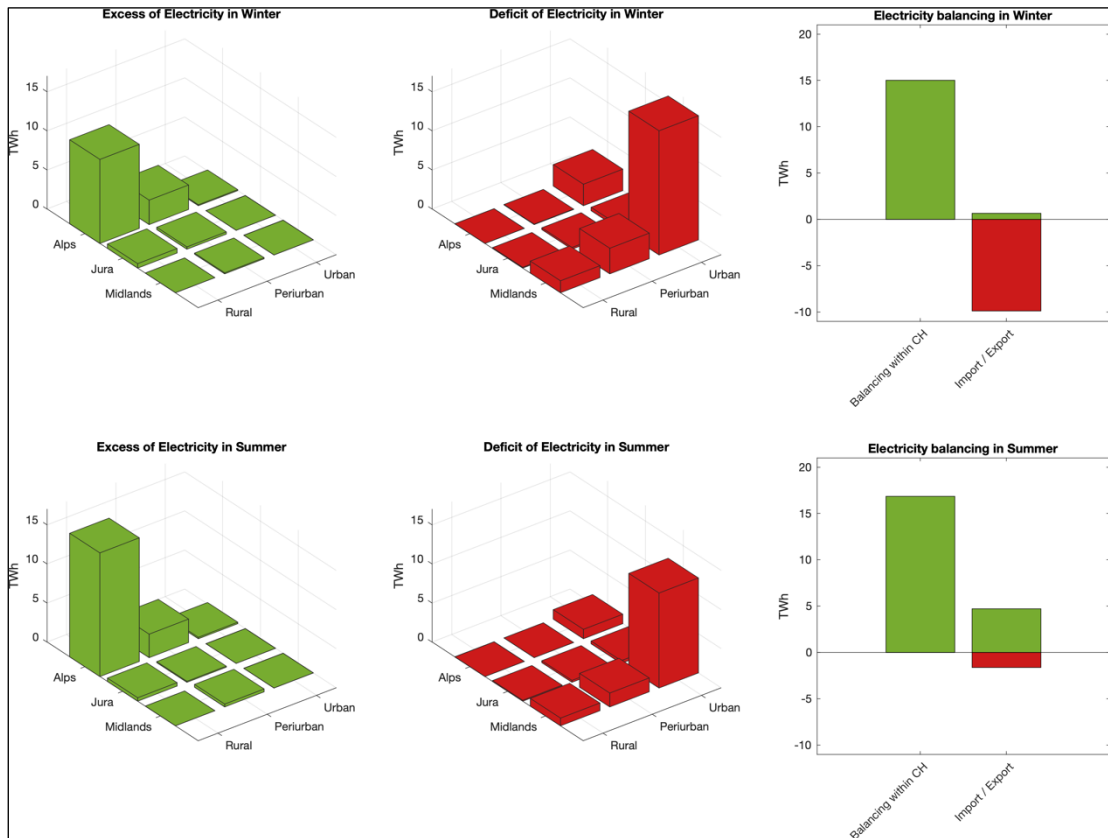


Figure 1. Balance of electricity from OREES in scenario A

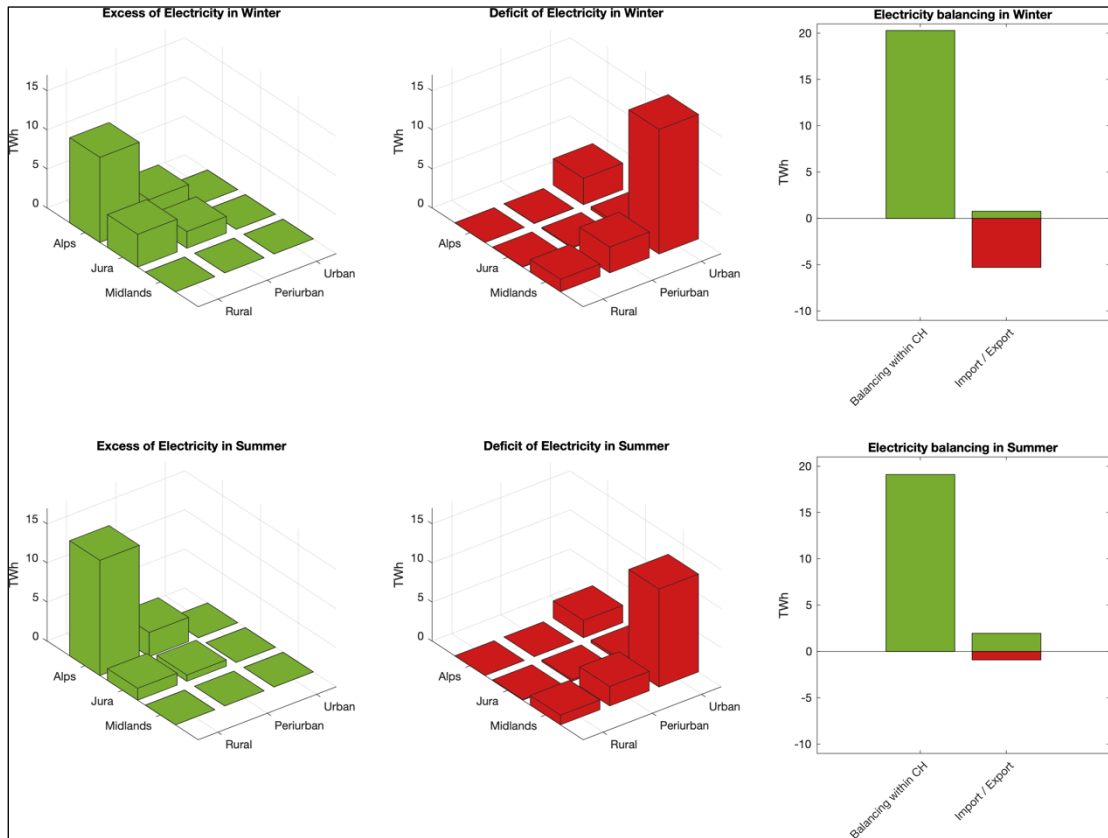


Figure 2. Balance of electricity from OREES in scenario B



3 Conclusion

The integration of wind and solar PV can now be assessed at a refined spatial level. This way, the local to regional production – demand dynamics can be investigated. The methodology provided here allows to adapt the installation criteria and optimization goals.

4 References

- Dujardin, J., A. Kahl, and M. Lehning. "Synergistic Optimization of Renewable Energy Installations through Evolution Strategy." *Environmental Research Letters* 16, no. 6 (June 2021). <https://doi.org/10.1088/1748-9326/abfc75>.
- Heinisch, Verena, Jérôme Dujardin, Paolo Gabrielli, Pranjal Jain, Michael Lehning, Giovanni Sansavini, Jan-Philipp Sasse, Christian Schaffner, Marius Schwarz, and Evelina Trutnevyte. "Inter-Comparison of Spatial Models for High Shares of Renewable Electricity in Switzerland." *Applied Energy* 350 (November 15, 2023): 121700. <https://doi.org/10.1016/j.apenergy.2023.121700>.